Mechanical Behavior of Materials (4th)

7.2 $\left.\begin{matrix}\sigma_1\\ \sigma_2\end{matrix}\right\} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$= \frac{125-80}{2} \pm \sqrt{\left(\frac{125+80}{2}\right)^2 + 35^2}$ MPa

$= \begin{cases} 129.3 \text{ MPa} \\ -84.3 \text{ MPa} \end{cases}$

$\sigma_3 = \sigma_z = 0$

$X = \frac{\sigma_{ut}}{\bar{\sigma}_{NT}} = \frac{\sigma_{ut}}{\max\{\sigma_1, \sigma_2, \sigma_3\}} = \frac{147 \text{ MPa}}{129.3 \text{ MPa}} = 1.14$

reference: textbook P281

7.6 $\left.\begin{matrix}\sigma_1\\ \sigma_2\end{matrix}\right\} = \frac{\sigma_1+\sigma_2}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$= \frac{365+138}{2} \pm \sqrt{\left(\frac{365-138}{2}\right)^2 + 69^2}$ MPa

$= \begin{cases} 385.9 \text{ MPa} \\ 117.1 \text{ MPa} \end{cases}$

$\sigma_3 = \sigma_z = -69$ MPa

(a) maximum shear stress criterion

$\bar{\sigma}_S = \max\{|\sigma_1-\sigma_2|, |\sigma_2-\sigma_3|, |\sigma_3-\sigma_1|\}$

$= |\sigma_3-\sigma_1| = 434.9$ MPa

$X_S = \frac{\sigma_0}{\bar{\sigma}_S} = \frac{1619 \text{ MPa}}{434.9 \text{ MPa}} = 3.72$

(b) octahedral shear stress criterion

$\bar{\sigma}_H = \frac{1}{\sqrt{2}}\sqrt{(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2} = 377.9$ MPa

$X_H = \frac{\sigma_0}{\bar{\sigma}_H} = 4.28$

7.11 For 7075-T6 Aluminum.

$E = 71$ GPa, $\sigma_0 = 469$ MPa, $\nu = 0.345$

$\begin{cases} \varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y) \\ \varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x) \\ \gamma_{xy} = \frac{1}{G}\tau_{xy} \end{cases}$

$\Rightarrow \begin{cases} \sigma_x = \frac{E}{(1-\nu^2)}(\varepsilon_x + \nu\varepsilon_y) = 294.3 \text{ MPa} \\ \sigma_y = \frac{E}{(1-\nu^2)}(\varepsilon_y + \nu\varepsilon_x) = 112.2 \text{ MPa} \\ \tau_{xy} = \frac{E}{2(1+\nu)}\gamma_{xy} = 18.5 \text{ MPa} \end{cases}$

$$\left.\begin{matrix}\sigma_1\\ \sigma_2\end{matrix}\right\} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2} = \begin{cases}286\,MPa\\ 110\,MPa\end{cases}$$

(a) maximum shear stress criterion

$$\bar{\sigma}_S = \max\{|\sigma_1|,\ |\sigma_2|,\ |\sigma_1-\sigma_2|\} = \sigma_1 = 286\,MPa$$

$$X_S = \frac{\sigma_0}{\bar{\sigma}_S} = 1.58$$

(b) octahedral shear stress criterion

$$\bar{\sigma}_H = \frac{1}{\sqrt{2}}\sqrt{(\sigma_1-\sigma_2)^2+\sigma_1^2+\sigma_2^2} = 250\,MPa$$

$$X_H = \frac{\sigma_0}{\bar{\sigma}_H} = 1.81$$

Answer: $X_S = 1.58$ or $X_H = 1.81$

7.28 (a) $\left.\begin{matrix}\sigma_1\\ \sigma_2\end{matrix}\right\} = -p \pm \sqrt{0^2+\tau_{xy}^2} = -p \pm |\tau_{xy}|$

According to the octahedral shear stress criterion, we have

$$\frac{\sigma_0}{X_H} = \frac{1}{\sqrt{2}}\sqrt{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2} = \sqrt{3}\,|\tau_{xy}|$$

So the largest value of τ_{xy} is

$$\tau_{xy} = \frac{\sigma_0}{\sqrt{3}X_H} = 108.3\,MPa$$

(b) In the equation $\frac{\sigma_0}{X_H} = \sqrt{3}\,|\tau_{xy}|$, there is no "p".

So actually, the effect of p is absent.

12.4. Equation 12.8 process omitted.

8.4. (a) steels AISI 1144,

$$a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_0}\right)^2 = \frac{1}{\pi}\times\left(\frac{66\,MPa\sqrt{m}}{540\,MPa}\right)^2 = 4.75\,mm$$

ASTM A517

$$a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_0}\right)^2 = \frac{1}{\pi}\times\left(\frac{187\,MPa\sqrt{m}}{760\,MPa}\right)^2 = 19.3\,mm$$

300-M (650°C temper)

$$a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_0}\right)^2 = \frac{1}{\pi}\times\left(\frac{152\,MPa\sqrt{m}}{1070\,MPa}\right)^2 = 6.42\,mm$$

300-M (300°C temper)

$$a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_0}\right)^2 = \frac{1}{\pi}\times\left(\frac{65\,MPa\sqrt{m}}{1700\,MPa}\right)^2 = 0.44\,mm$$

aluminum alloys 2219-T851

$$a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_0}\right)^2 = \frac{1}{\pi}\times\left(\frac{36\,MPa\sqrt{m}}{350\,MPa}\right)^2 = 3.37\,mm$$

7075-T651

$$a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_0}\right)^2 = \frac{1}{\pi}\times\left(\frac{24MPa\sqrt{m}}{505MPa}\right)^2 = 1.05mm$$

polymers ABS

$$a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_0}\right)^2 = 1.15mm - 2.48mm$$

epoxy $a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_f}\right)^2 = 0.014mm - 0.15mm$

soda-lime glass $a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_u}\right)^2 = \frac{1}{\pi}\times\left(\frac{0.76MPa\sqrt{m}}{50MPa}\right)^2 = 0.074mm$

SiC $a_t = \frac{1}{\pi}\left(\frac{K_c}{\sigma_u}\right)^2 = \frac{1}{\pi}\times\left(\frac{3.7MPa\sqrt{m}}{307MPa}\right)^2 = 0.046mm$

(b) brittle materials.

8.10 (a) $\frac{a}{b} = \frac{10}{50} = 0.2 > 0.13$

so $F = 0.265(1-0.2)^4 + \frac{0.857+0.265\times 0.2}{(1-0.2)^{3/2}} = 1.38$

$$K = \frac{K_{IC}}{X_K} = \frac{24MPa\sqrt{m}}{3} = 8MPa\sqrt{m}$$

$$S_{g\,max} = \frac{K}{F\sqrt{\pi a}} = \frac{8MPa\sqrt{m}}{1.38\times\sqrt{\pi\times 10mm}} = 32.7MPa$$

$$P_{max} = S_{g\,max}bt = 32.7MPa\times 50mm\times 5mm = 8.175kN$$

(b) $S_g = \frac{P}{bt} = \frac{5kN}{50mm\times 5mm} = 20MPa$

if $a/b \le 0.13$

$$a_{max} = \frac{1}{\pi}\left(\frac{K}{FS_g}\right)^2 = 40.6mm$$

obviously $a/b > 0.13$

Then we can obtain the equation:

$$K = \left[0.265(1-\alpha)^4 + \frac{0.857+0.265\alpha}{(1-\alpha)^{3/2}}\right]S_g\sqrt{\pi a}$$

solve the equation and we can yield.

$a_{max} = 16.4mm$

8.15 (a) $S_g = \frac{6M}{b^2t} = \frac{6\times 5kN\cdot m}{(40mm)^2\times 10mm} = 1875MPa$

$$K = 1.12S_g\sqrt{\pi a} = 117.7MPa\sqrt{m}$$

$$X_K = \frac{K_c}{K} = \frac{176MPa\sqrt{m}}{117.7MPa\sqrt{m}} = 1.49$$

(b) $\frac{a}{b} = 0.25 \le 0.4$

so $F = 1.12$.

$$K = 1.12S_g\sqrt{\pi a} = 372.2MPa\sqrt{m}$$

$$X_K = \frac{K_c}{K} = \frac{176MPa\sqrt{m}}{372.2MPa\sqrt{m}} < 1$$

It is very dangerous and we should replace it.

8.47 (a) $K = 48.2\ MPa\sqrt{m}$

According to (8.40). the overall requirement for plane strain is

$$t,\ a,\ (b-a),\ h \geq 2.5\left(\frac{K}{\sigma_0}\right)^2 = 10.5\,mm$$

According to (8.38), an overall limit on the use of LEFM is

$$a,\ (b-a),\ h \geq \frac{4}{\pi}\left(\frac{K}{\sigma_0}\right)^2 = 5.3\,mm$$

All satisfied, so LEFM applicable.

(b) $2r_{o\sigma} = \frac{1}{\pi}\left(\frac{K}{\sigma_0}\right)^2 = \frac{1}{\pi}\times\left(\frac{48.2\ MPa\sqrt{m}}{760\,MPa}\right)^2$

$= 1.33mm$

9.10 (a) Applying $\sigma_a = AN_f^B$. we can get

$$\sigma_{a1} = AN_f^B = 1888\,MPa \times 60000^{-0.104} = 601.6\,MPa$$

$$X_S = \frac{\sigma_{a1}}{\sigma_a} = \frac{601.6\,MPa}{500\,MPa} = 1.2$$

$$N_{f2} = \left(\frac{\sigma_a}{A}\right)^{\frac{1}{B}} = \left(\frac{500\,MPa}{1888\,MPa}\right)^{-\frac{1}{0.104}} = 355300$$

$$X_N = \frac{N_{f2}}{N_f} = \frac{355300}{60000} = 5.92$$

Safety factors in the stress for fatigue, should be similar in magnitude to other stress-based safety factors, typically in the range $X_S = 1.5$ to 3.0

(reference: textbook P427)

Therefore, for an actual engineering application, we usually take a larger safety factor in stress.

Safety factor in life need to be in the range $X_N = 5$ to 20 or more, so X_N in this question seems reasonable for an actual engineering application.

9.11 (a) From the equation $\sigma_{a1} = AN_f^B$, we can get

$$N_f = \left(\frac{X_s \sigma_a}{A}\right)^{\frac{1}{B}} = \left(\frac{1.5\times 150\,MPa}{1006\,MPa}\right)^{-\frac{1}{0.115}} = 4.53\times10^5$$

(b) From the equation $\sigma_a = AN_{f2}^B$. we can get

$$N_{f2} = \left(\frac{\sigma_a}{A}\right)^{\frac{1}{B}} = \left(\frac{150\,MPa}{1006\,MPa}\right)^{-\frac{1}{0.115}} = 1.54\times10^7$$

Therefore, the corresponding safety factor on life is $X_N = \frac{N_{f2}}{N_f} = \frac{1.54\times10^7}{4.53\times10^5} = 34$

9.21 (a) $$N_f = \left(\frac{\sigma_a}{A}\right)^{\frac{1}{B}} = \left(\frac{200\,MPa}{927\,MPa}\right)^{-\frac{1}{0.138}} = 6.7\times10^4$$

(b) According to the Morrow equation $\frac{\sigma_a}{\sigma_{ar}} + \frac{\sigma_m}{\sigma_f'} = 1$

$$\sigma_{ar} = \frac{\sigma_a}{1-\frac{\sigma_m}{\sigma_f'}} = \frac{200\,MPa}{1-\frac{80\,MPa}{1020\,MPa}} = 217\,MPa$$

$$N_f = \left(\frac{\sigma_{ar}}{A}\right)^{\frac{1}{B}} = \left(\frac{217\,MPa}{927\,MPa}\right)^{-\frac{1}{0.138}} = 3.7\times10^4$$

(c) $$\sigma_{ar} = \frac{\sigma_a}{1-\frac{\sigma_m}{\sigma_f'}} = \frac{200\,MPa}{1-\frac{-80\,MPa}{1020\,MPa}} = 185.5\,MPa$$

$$N_f = \left(\frac{\sigma_{ar}}{A}\right)^{\frac{1}{B}} = \left(\frac{185.5\,MPa}{927\,MPa}\right)^{-\frac{1}{0.128}} = 1.16\times10^5$$

9.37. $$\sigma_{ar1} = AN_f^B = 839\times5000^{-0.102}\,MPa = 352\,MPa$$

Morrow equation:

$$\frac{\sigma_a}{\sigma_{ar}} + \frac{\sigma_m}{\sigma_f'} = 1$$

$$\sigma_{ar} = \frac{\sigma_a}{1-\frac{\sigma_m}{\sigma_f'}} = \frac{200\,MPa}{1-\frac{250\,MPa}{900\,MPa}} = 277\,MPa$$

$$X_s = \frac{\sigma_{ar1}}{\sigma_{ar}} = \frac{352\,MPa}{277\,MPa} = 1.27$$

$$N_{f2} = \left(\frac{\sigma_{ar}}{A}\right)^{\frac{1}{B}} = \left(\frac{277\,MPa}{839\,MPa}\right)^{-\frac{1}{0.102}} = 5.2\times10^4$$

$$X_N = \frac{N_{f2}}{N_f} = \frac{5.2\times10^6}{5000} = 10.5$$